



The following programme of work is designed to ensure that the students are prepared to attempt the Mathematical Methods and the Specialist Mathematics courses in Year 11.

Weeks	WA Curriculum Descriptor UNIT TITLE	Learning Objectives	Extension, Enrichment	Skills/General Capabilities	Assessment	Resources
Term 1 1-5	Algebra	- Review solving and graphing quadratics - Identify the leading term, degree, coefficients, and constants of a polynomial - Factorise a polynomial by polynomial division - Apply the factor remainder theorem - Review solving simultaneous equations	Mars Collaborative Project	Persisting Thinking and communicating with clarity and precision Thinking flexibly Striving for accuracy Questioning and posing problems Applying past knowledge to new situations	Week 5 Test 1	CAM: 3A,3C,3D,3E 6A-D <i>12A-E</i> ICE10B1: 5A-5G ICE10B2: 18A-F

Resources:

CAM – Cambridge Additional Mathematics
 ICE10B1 – ICE-EM Mathematics Year 10 Book 1
 ICE10B2- ICE-EM Mathematics Year 10 Book 2

	and Exams					
7-9	Trigonometric Functions	- Define trigonometric ratios - Identify amplitude (undefined for tan), period and principal axis of sine, cosine and tangent graphs - Describe and graph transformations of sine, cosine, tangent functions - Model periodic phenomena using trigonometric functions		Persisting Thinking and communicating with clarity and precision Thinking flexibly Striving for accuracy Questioning and posing problems Applying past knowledge to new situations		CAM 9A-D
Term 3 1-4	Circle Geometry	- Define and use radian measure - Calculate lengths of arcs and areas of sectors and segments using radian measure - Apply angle and chord properties of a circle - Solve problems involving latitude and longitude - Use and apply circle theorems		Persisting Thinking and communicating with clarity and precision Thinking flexibly Striving for accuracy Questioning and posing problems Applying past knowledge to new situations	Week 4 Test 4	CAM: 8A-C ICE10B2: 14A-F

5-10	Counting and Probability	• Use Venn diagrams, tree diagrams and probability tables to calculate probabilities • Use and apply basic concepts and rules of probability • Recognise and solve problems which involve conditional probability • Recognise and justify mutually exclusive events • Recognise and justify independent events • Use the multiplication rule and the law of total probability to calculate probabilities • Apply the addition and multiplication principle • Solve problems which involve Permutations and Combinations • Apply counting methods to probability		Persisting Thinking and communicating with clarity and precision Thinking flexibly Striving for accuracy Questioning and posing problems Applying past knowledge to new situations	Week 10 Test 5	CAM: 10A-E ICE10B2: 13A-D 16A-D
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Term 4 1-3	Exponentials and Logarithms	- Solve exponential equations - Solve growth and decay problems - Use the definition of a logarithm to establish and apply the laws of logarithms to convert between exponential and logarithmic forms - Evaluate logarithms - Simplify expressions using logarithm laws	Owing Home Project	a Persisting Thinking and communicating with clarity and precision Thinking flexibly Striving for accuracy Questioning and posing problems Applying past knowledge to new situations	Week 2 EPW2	CAM: 4B,C,E,F 5A-G ICE10B1 9E,G ICE10B2 15A-E
4-5	Vectors	<i>Define and Identify examples of vectors</i> <i>Use a variety of notations and representations for vectors in two dimensions</i> <i>Define and use the magnitude and direction of a vector</i> <i>Perform addition and subtraction of vectors and multiplication of a vector by a scalar algebraically and geometrically and interpret these in geometric terms.</i>		Persisting Thinking and communicating with clarity and precision Thinking flexibly Striving for accuracy Questioning and posing problems Applying past knowledge to new situations		CAM: 11A-G
6-7	Consolidation and Exams					
8-10	Rates of Change	- Distinguish between instantaneous and average rate of change - Interpret the difference quotient as the average rate of change of a function - Investigate the instantaneous rate of change at a point				CAM: 13A-C

ASSESSMENT OUTLINE

SEMESTER 1

Assessment Item	Weighting	Timing
Test 1	15%	Term 1 Week 5
Test 2	15%	Term 1 Week 11
Test 3	15%	Term 2 Week 5
EPW 1	15%	Term 1 Week 8
Semester One Exam	40%	Term 2 Week 5/6

SEMESTER 2

Assessment Item	Weighting	Timing
Test 4	15%	Term 3 Week 4
Test 5	15%	Term 3 Week 10
Test 6	15%	Term 4 Week 5
EPW 2	15%	Term 4 Week 2
Semester Two Exam	40%	Term 4 Week 6/7